

Analog Drum Synthesizer Circuit

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May 2026

Abstract

This paper presents the theoretical design, mathematical derivation, and implementation of an FM Drum Synthesizer, inspired by the Roland TR-909. The modular architecture includes a gate-to-trigger converter, pitch envelope, voltage-controlled oscillator (VCO), diode wave shaper, voltage-controlled attenuator (VCA), and decay envelope. Through the generation of continuous triangle waves and subsequent subtractive shaping into pseudo-sines, coupled with accurate VCA envelope modulation, the circuit effectively emulates percussive timbres like kick drums, toms, and snares.

1. Introduction

This project documents a hardware FM drum synthesizer circuit built and tested as part of EW-2, EC2.202. The objective is to design and implement the core analog synthesis engine inspired by the iconic Roland TR-909 drum machine from 1984.

Historically, the TR-808 used highly efficient bridged-T oscillators capable of producing percussive sine wave hits with minimal components. In contrast, the TR-909 utilized full continuous voltage-controlled oscillators (VCOs) that generate triangle waves. This shift traded efficiency for flexibility, allowing for advanced subtractive synthesis and complex wave shaping. Furthermore, historical documentation and modern analysis show the original TR-909 was actually a hybrid machine: it utilized 6-bit PCM samples for cymbals and hi-hats, but engineered purely analog, discrete component circuits for its kick, snare, and toms to retain maximum low-end punch and

tweakability. By exploiting this continuous analog waveform architecture, our design extends the classic 909 kick drum topology into a broader foundational engine capable of generating kick drums, toms, and foundational FM percussive textures.

2. Design Specifications and Challenges

The system was designed to meet several core goals while overcoming hardware realities:

- **Oscillation:** Voltage-controlled continuous triangle/sine generation across a wide frequency range.
- **Shaping:** Percussive amplitude shaping using a dedicated VCA and exponential decay envelope.
- **Transients:** Consistent attack transients achieved via a VCO capacitor-bridge reset

mechanism, eliminating random attack clicks.

- **Modulation:** A 909-style pitch bend (pitch envelope) delivering deep analog "thump."
- **Hardware Constraints:** Addressing VCA DC offset drift through manual trimmer calibration and compensating for thermal drift in discrete transistor configurations.

3. System Overview

3.1 Signal-Flow Architecture

The system architecture is broken down into several modular stages, each fulfilling a specific functional role. A gate-to-trigger converter fires short trigger spikes based on sequencer inputs, which initiate tandem pitch and decay envelopes. The continuous triangle VCO feeds a diode wave-shaper, which is then fed into a discrete VCA modulated by the decay envelope.

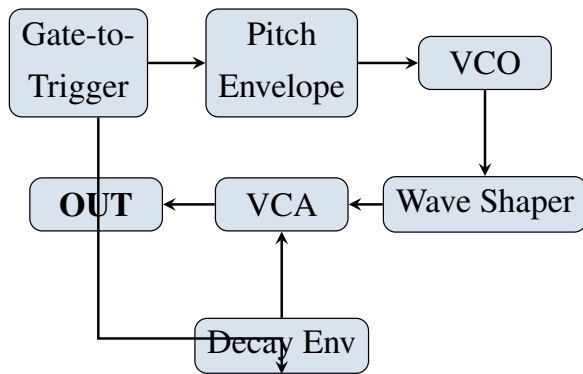


Figure 1: Signal-flow block diagram

4. Stage 1: Voltage Controlled Oscillator (VCO)

4.1 Design Rationale

The Voltage Controlled Oscillator (VCO) forms the fundamental sound-generation stage of the analog drum synthesizer. Its primary purpose is to

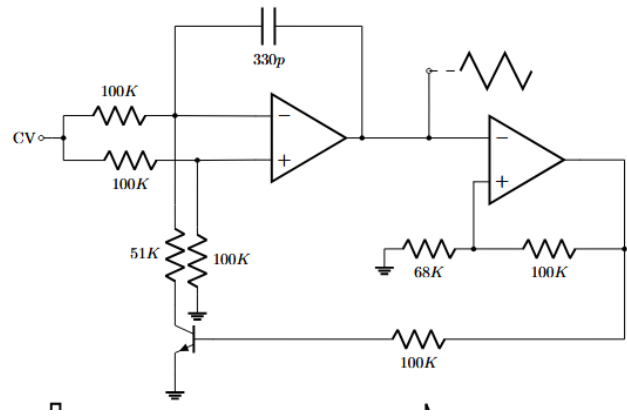


Figure 2

generate a stable periodic waveform whose frequency can be dynamically varied using an external control voltage. In this project, a triangular waveform was selected because of its smooth harmonic profile and its suitability for further analog wave-shaping into drum-like tones.

To achieve stable oscillation with controllable frequency, the VCO was implemented using an integrator cascaded with a Schmitt trigger configuration. The integrator generates a linear voltage ramp, while the Schmitt trigger continuously switches polarity whenever predefined voltage thresholds are crossed. This feedback mechanism creates sustained oscillations.

The oscillator frequency is directly controlled by the applied control voltage (V_{CV}). By varying this voltage, the charging and discharging rate of the integrator capacitor changes, producing different frequencies. This property becomes especially important in percussion synthesis, where rapidly changing pitch envelopes create realistic drum transients and attack characteristics.

The circuit was designed using OP07 operational amplifiers powered by ± 12 V rails to ensure sufficient output swing and stable analog operation over the required audio frequency range.

4.2 Mathematical Derivation

The VCO consists of an OP07 integrator (U1) cascaded with an OP07 Schmitt trigger (U2). The integrator converts a constant input current into a linear voltage ramp, while the Schmitt trigger provides hysteresis-based switching to sustain oscillation.

Component values:

- $R_2 = R_3 = 100 \text{ k}\Omega$
- $C_1 = 330 \text{ pF}$
- $R_1 = 68 \text{ k}\Omega$
- $R_7 = 100 \text{ k}\Omega$
- $R_4 = 100 \text{ k}\Omega$
- $R_5 = 51 \text{ k}\Omega$

The integrator operation follows the capacitor current relation:

$$I_C = C_1 \frac{dV_{out}}{dt} \quad (1)$$

The current entering through resistor R_2 is:

$$I_{in} = \frac{V_{CV}}{R_2} \quad (2)$$

Assuming ideal op-amp operation, the inverting terminal remains at virtual ground. Therefore:

$$I_{C1} = -I_{in} \quad (3)$$

Substituting:

$$C_1 \frac{dV_{out}}{dt} = -\frac{V_{CV}}{R_2} \quad (4)$$

Rearranging:

$$\frac{dV_{out}}{dt} = -\frac{V_{CV}}{R_2 C_1} \quad (5)$$

Integrating both sides:

$$V_{out}(t) = -\frac{1}{R_2 C_1} \int V_{CV} dt + V_{initial} \quad (6)$$

For a constant control voltage, the output becomes a linear voltage ramp. Substituting the component values:

$$\frac{dV_{out}}{dt} = -\frac{V_{CV}}{(100 \times 10^3)(330 \times 10^{-12})} \quad (7)$$

$$\frac{dV_{out}}{dt} = -3.03 \times 10^4 V_{CV} \text{ V/s} \quad (8)$$

This shows that the slope of the triangular waveform is directly proportional to the applied control voltage. Increasing V_{CV} causes the capacitor to charge faster, thereby increasing the oscillation frequency.

The Schmitt trigger establishes upper and lower switching thresholds through positive feedback:

$$\frac{V_{TH}}{V_{sat}} = \frac{R_1}{R_1 + R_7} \quad (9)$$

Substituting values:

$$\frac{V_{TH}}{11} = \frac{68k}{68k + 100k} \quad (10)$$

$$V_{TH} \approx \pm 4.44 \text{ V} \quad (11)$$

The integrator output ramps between these thresholds. Since the ramp traverses $2V_{TH}$ during each half-cycle:

$$T_{half} = \frac{2V_{TH} R_2 C_1}{V_{CV}} \quad (12)$$

The full oscillation period becomes:

$$T = 2T_{half} = \frac{4V_{TH} R_2 C_1}{V_{CV}} \quad (13)$$

Therefore, the oscillation frequency is:

$$f_0 = \frac{V_{CV}}{4R_2C_1V_{TH}} \quad (14)$$

Substituting the designed values:

$$f_0 = \frac{V_{CV}}{5.86 \times 10^{-4}} \quad (15)$$

For a tuning control voltage of $V_{CV} = 3 \text{ V}$:

$$f_0 \approx 5120 \text{ Hz} \quad (16)$$

Thus, the oscillator successfully operates within the intended audio frequency range suitable for analog percussion synthesis.

The slope equation obtained from the integrator is:

$$\frac{dV_{out}}{dt} = -3.03 \times 10^4 V_{CV} \text{ V/s} \quad (17)$$

Physical Meaning:

The above expression shows that the rate of change of the triangular waveform is directly controlled by the input control voltage V_{CV} . Physically, the control voltage determines the current flowing into the integrator capacitor. Since the capacitor charging current increases with increasing V_{CV} , the capacitor charges and discharges more rapidly, producing a steeper voltage ramp.

The slope is also inversely proportional to the RC time constant (R_2C_1). Larger resistor or capacitor values slow down the charging process, resulting in lower oscillation frequencies, while smaller values increase the charging speed and raise the oscillation frequency.

Thus, the VCO converts variations in control voltage directly into variations in frequency. This property is essential for percussion synthesis because it allows rapid pitch sweeps and transient frequency modulation, which are responsible for creating realistic drum attack characteristics.

4.3 Results and Interpretation

The VCO was successfully implemented and tested in both simulation and hardware environments. The generated output waveform was a stable triangular wave whose frequency varied proportionally with the applied control voltage.

The LTspice simulation results verified the theoretical derivations. As the control voltage increased, the slope of the integrator ramp increased correspondingly, resulting in higher oscillation frequencies. Lower control voltages produced slower charging and discharging cycles, generating lower-frequency triangular waves.

The chosen component values provided a sufficiently wide frequency sweep suitable for drum synthesis applications. The integrator capacitor value of 330 pF enabled rapid waveform transitions, while the resistor values established stable operation without excessive noise sensitivity.

The Schmitt trigger thresholds remained highly stable throughout operation, ensuring clean waveform transitions and preventing unwanted oscillation jitter. The positive feedback network successfully created hysteresis, eliminating false switching due to small voltage fluctuations or noise.

An important practical addition to the circuit was the inclusion of transistor Q1 (BC547B), which acts as a hard-reset mechanism for the timing capacitor during trigger events. This ensured that each drum transient began from a fixed oscillator phase, thereby reducing random phase inconsistencies and minimizing audible clicking artifacts during playback.

Minor differences between theoretical and measured frequencies were observed. These deviations were primarily caused by:

- tolerance variations in resistors and capacitors,
- finite slew rate limitations of the OP07,

- parasitic capacitances from breadboard wiring,
- transistor leakage currents,
- and non-ideal op-amp saturation behavior.

Despite these practical non-idealities, the oscillator exhibited highly stable and repeatable behavior. The waveform quality and controllable frequency sweep made the VCO suitable for subsequent drum synthesis stages such as wave shaping and pitch modulation.

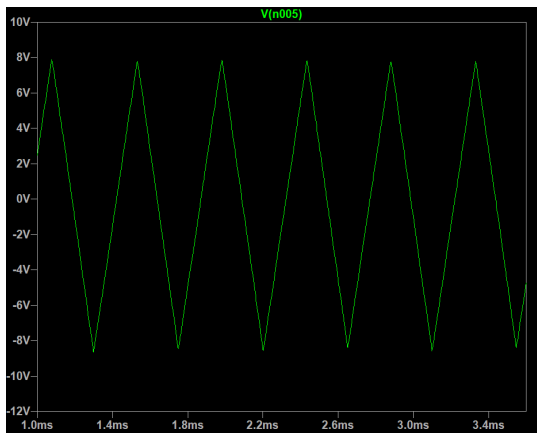


Figure 3: LTspice transient simulation showing triangular waveform generation for different control voltages.

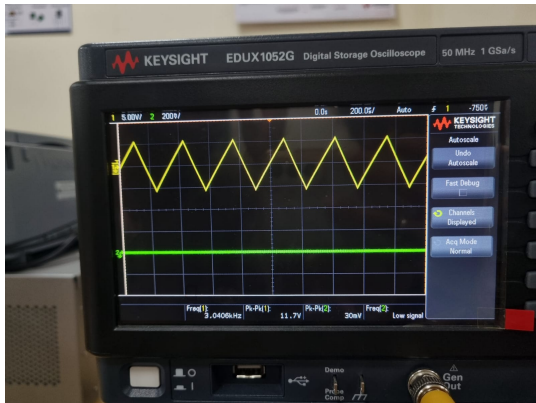


Figure 4: Oscilloscope capture of the implemented VCO triangular waveform.

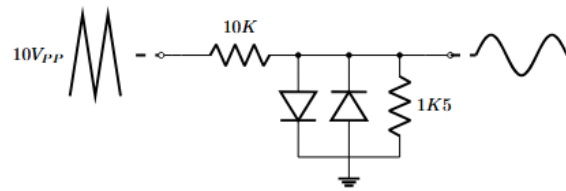


Figure 5: Anti-parallel diode wave shaper used for converting the triangular waveform into a pseudo-sinusoidal waveform.

5. Stage 2: Diode Wave Shaper

5.1 Design Rationale

The raw triangular waveform generated by the VCO contains strong odd harmonics and sharp linear peaks. While this waveform is useful for synthesis, it does not closely resemble the smooth low-frequency tonal characteristics typically associated with analog kick drums. Therefore, a wave-shaping stage was introduced to transform the triangular waveform into a smoother pseudo-sinusoidal waveform.

The wave shaper operates using the non-linear current-voltage characteristics of semiconductor diodes. By placing two diodes in an anti-parallel configuration, the positive and negative peaks of the triangular waveform are compressed symmetrically. This gradual compression rounds the sharp corners of the waveform and reduces higher harmonic content, producing a waveform that more closely approximates a sine wave.

Unlike hard clipping circuits that abruptly limit the waveform and introduce harsh harmonics, the diode-based wave shaper performs soft non-linear compression, resulting in a smoother and more natural analog drum tone.

5.2 Mathematical Derivation

The wave shaper circuit consists of two 1N4148 switching diodes connected in an anti-parallel configuration along with a series limiting resistor:

- $R_8 = 10 \text{ k}\Omega$
- $D_1, D_2 = 1\text{N4148}$

The current flowing through each diode follows the Shockley diode equation.

For the positive half-cycle:

$$I_{D1} = I_S (e^{V_{out}/\eta V_T} - 1) \quad (18)$$

For the negative half-cycle:

$$I_{D2} = I_S (e^{-V_{out}/\eta V_T} - 1) \quad (19)$$

where:

- I_S = reverse saturation current
- $V_T \approx 26 \text{ mV}$ = thermal voltage
- η = diode ideality factor

Combining the two currents:

$$I_{D1} + I_{D2} = 2I_S \sinh\left(\frac{V_{out}}{\eta V_T}\right) \quad (20)$$

Applying Kirchhoff's Current Law at the output node:

$$\frac{V_{\Delta} - V_{out}}{R_8} = 2I_S \sinh\left(\frac{V_{out}}{\eta V_T}\right) \quad (21)$$

$$\frac{V_{\Delta} - V_{out}}{R_8} = 2I_S \sinh\left(\frac{V_{out}}{\eta V_T}\right) \quad (22)$$

For small signal amplitudes where:

$$|V_{out}| \ll \eta V_T \quad (23)$$

the hyperbolic sine function behaves approximately linearly:

$$\sinh(x) \approx x \quad (24)$$

and therefore the waveform passes through with minimal distortion.

However, when the waveform amplitude approaches the diode forward voltage threshold ($\approx$

0.6 V), exponential conduction begins. The diodes conduct heavily near the waveform peaks, effectively compressing the sharp triangular edges and rounding the waveform into a pseudo-sinusoidal shape.

Physical Meaning:

The anti-parallel diode arrangement behaves as a dynamic non-linear resistance. At low voltages, the diodes remain mostly OFF and the triangular waveform passes unchanged. As the voltage increases toward the diode threshold, the diodes begin conducting exponentially, reducing the rate at which the output voltage can increase.

This selective peak compression smoothens the sharp edges of the triangular waveform. Since high-frequency harmonics are mainly generated by sharp waveform transitions, reducing these sharp transitions suppresses higher harmonics and produces a waveform that more closely resembles a sine wave.

Thus, the wave shaper acts as a passive analog harmonic reduction stage that improves the tonal quality of the synthesized drum sound.

5.3 Results and Interpretation

The wave shaper successfully transformed the incoming triangular waveform into a smoother pseudo-sinusoidal waveform suitable for analog drum synthesis applications.

Both LTspice simulations and practical oscilloscope measurements confirmed that the anti-parallel diode arrangement effectively rounded the sharp peaks of the triangular waveform. At lower signal amplitudes, minimal waveform distortion was observed because the diodes remained below their forward conduction threshold. As the signal amplitude increased, the diodes entered exponential conduction and progressively compressed the waveform peaks.

The resistor value $R_8 = 10 \text{ k}\Omega$ played an im-

portant role in controlling the shaping strength and limiting excessive diode current. A smaller resistor value would produce stronger clipping and increased harmonic distortion, while a larger resistor value would weaken the shaping effect and preserve more triangular characteristics.

The resulting waveform exhibited significantly reduced high-frequency harmonic content compared to the original triangular wave. This reduction in harmonics produced a smoother and warmer tone that more closely resembled the low-frequency body of analog kick drums.

Unlike hard clipping methods that produce abrupt saturation and harsh overtones, the soft non-linear shaping provided by the diodes generated a more natural analog sound character. This made the circuit particularly effective for percussion synthesis applications where smooth transient behavior is desirable.

Minor asymmetries were observed in the practical waveform due to:

- diode forward voltage mismatches,
- temperature-dependent diode characteristics,
- parasitic wiring effects,
- and slight variations in component tolerances.

Despite these non-idealities, the implemented circuit produced stable and repeatable waveform shaping with good agreement between theoretical predictions, LTspice simulations, and hardware measurements.

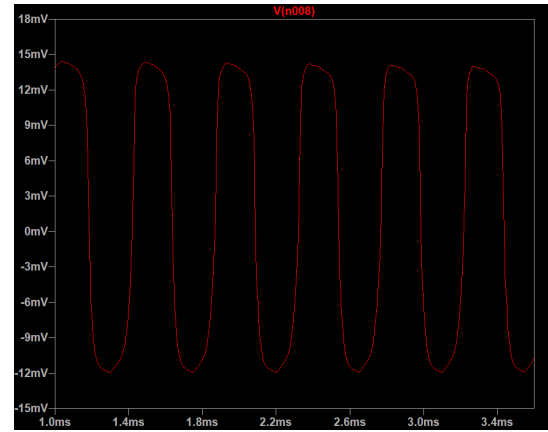


Figure 6: LTspice simulation showing triangular-to-pseudo-sine waveform conversion using the diode wave shaper.

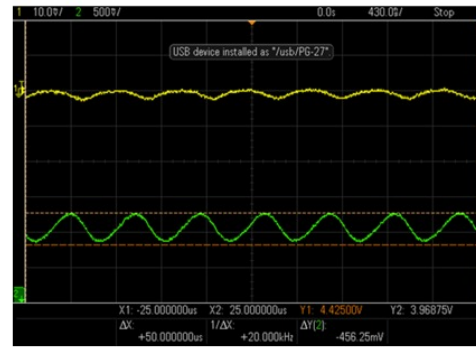


Figure 7: Oscilloscope capture of the practical pseudo-sinusoidal output waveform generated by the diode wave shaper.

6. Stage 3: Voltage Controlled Attenuator (VCA) and Decay Envelope

6.1 Design Rationale

One of the most important characteristics of a drum sound is its transient behavior. A typical drum signal consists of a very fast initial attack followed by a gradual exponential decay in amplitude. Reproducing this behavior electronically requires controlling the amplitude of the generated waveform over time.

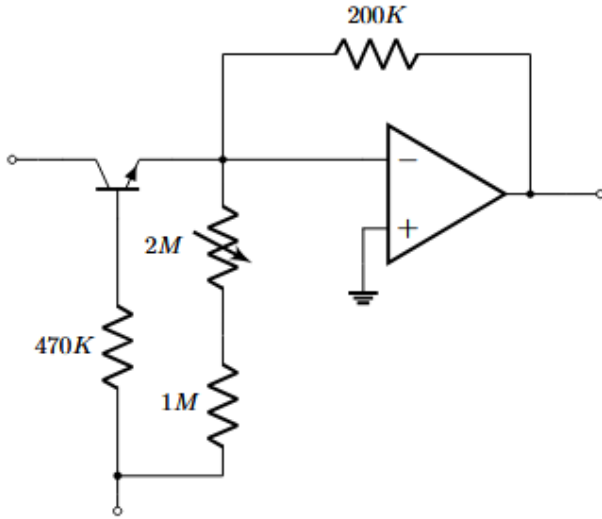


Figure 8: VCA Circuit

To implement this functionality, a Voltage Controlled Attenuator (VCA) was designed using a discrete transistor-based approach instead of dedicated VCA integrated circuits. The objective was to achieve analog amplitude modulation using minimal components while maintaining low cost and educational transparency.

The VCA works by exploiting the small-signal transconductance behavior of a bipolar junction transistor (BJT). The incoming pseudo-sinusoidal waveform from the wave shaper is first attenuated to the millivolt range using a resistive voltage divider. This small signal is then applied to the transistor operating in its active region.

Simultaneously, a decay envelope circuit generates an exponentially decaying control voltage using an RC discharge network. This decaying voltage is applied to the transistor base-emitter junction, dynamically controlling the transistor transconductance over time.

As a result, the transistor behaves as a voltage-controlled gain element whose gain decreases exponentially with time. Therefore, the output waveform becomes an exponentially decaying sinusoid of the form:

$$e^{-t/\tau} \sin(\omega t) \quad (25)$$

This waveform closely resembles the natural decay behavior of analog percussion instruments such as kick drums and toms.

6.2 Mathematical Derivation of the VCA

The VCA uses transistor Q2 (BC547B) as a voltage-controlled transconductance element in conjunction with OP07 op-amp U3.

The pseudo-sinusoidal waveform generated from the wave shaper is first attenuated using the resistor divider network:

- $R_9 = 2\text{ M}\Omega$
- $R_{10} = 47\text{ k}\Omega$

The attenuated signal becomes:

$$V_{sine,mV} = V_{\sim} \cdot \frac{R_{10}}{R_9 + R_{10}} \quad (26)$$

Substituting values:

$$V_{sine,mV} = V_{\sim} \cdot \frac{47}{2047} \approx 0.023V_{\sim} \quad (27)$$

Thus, the waveform amplitude is reduced to the millivolt range suitable for small-signal transistor operation.

The collector current generated by the transistor is approximately:

$$I_C(t) \approx \frac{V_{sine,mV}}{R_{10}} \sin(\omega t) \quad (28)$$

For a BJT operating in the active region, the small-signal transconductance is:

$$g_m(t) = \frac{I_C(t)}{V_T} \quad (29)$$

where:

- $V_T \approx 26 \text{ mV}$ is the thermal voltage

The decay envelope is generated using capacitor C_2 and the RC discharge network:

- $C_2 = 1 \mu\text{F}$
- $R_{14} = 1 \text{ k}\Omega$
- $R_{15} = 4.7 \text{ k}\Omega$ (variable decay control)

The envelope voltage follows the exponential discharge relation:

$$v_{be}(t) = V_0 e^{-t/\tau_{decay}} \quad (30)$$

where the decay time constant is:

$$\tau_{decay} = C_2(R_{14} + R_{15}) \quad (31)$$

Substituting values:

$$\tau_{decay} = (1 \mu\text{F})(1 \text{ k} + 4.7 \text{ k}) = 5.7 \text{ ms} \quad (32)$$

The transistor collector current therefore becomes:

$$i_c(t) = g_m(t) \cdot v_{be}(t) \quad (33)$$

Substituting for $g_m(t)$:

$$i_c(t) = \frac{I_C(t)}{V_T} \cdot V_0 e^{-t/\tau_{decay}} \quad (34)$$

The op-amp output voltage is related through feedback resistor R_{12} :

$$V_{out}(t) = -R_{12}i_c(t) \quad (35)$$

Substituting all terms:

$$V_{out}(t) = -\frac{R_{12}V_0}{V_T R_{10}} \cdot V_{sine} \cdot e^{-t/\tau_{decay}} \sin(\omega t) \quad (36)$$

Physical Meaning:

The output waveform consists of a sinusoidal carrier multiplied by an exponentially decaying envelope. Initially, the envelope voltage is maximum, resulting in large transistor transconductance and therefore high output amplitude. As time progresses, the RC network discharges exponentially, reducing the base-emitter control voltage and lowering the transistor gain.

This causes the sinusoidal waveform amplitude to decay exponentially over time, reproducing the natural decay characteristics of percussion instruments.

The decay rate is directly controlled by the RC time constant. Increasing resistance R_{15} increases the discharge time, producing longer drum tails suitable for tom-like sounds. Reducing the resistance produces rapid decay, resulting in short punchy kick drum sounds.

Thus, the VCA and decay envelope together form the primary amplitude-shaping mechanism of the drum synthesizer.

6.3 Results and Interpretation

The VCA and decay envelope stage successfully generated exponentially decaying sinusoidal waveforms characteristic of analog percussion instruments.

Both LTspice simulations and practical hardware measurements confirmed that the circuit produced the expected amplitude modulation behavior. Immediately after triggering, the output waveform exhibited maximum amplitude corresponding to the fully charged envelope capacitor. As the capacitor discharged through the RC network, the waveform amplitude decreased exponentially over time.

The decay response closely followed the theoretical envelope equation:

$$e^{-t/\tau_{decay}} \quad (37)$$

The potentiometer R_{15} effectively controlled the decay time constant and therefore allowed real-time tuning of the drum sound characteristics. Lower resistance settings produced short and punchy decay profiles suitable for kick drums, while higher resistance values produced longer resonant decays similar to tom or bass percussion sounds.

The resistor divider formed by R_9 and R_{10} was critical in ensuring proper small-signal transistor operation. Without sufficient attenuation, the transistor would enter non-linear saturation regions and produce distorted output waveforms.

The transistor-based VCA demonstrated effective analog gain control despite using a very small number of components. The op-amp feedback network amplified the controlled collector current into a usable audio signal while maintaining stable operation.

A practical challenge observed during implementation was DC offset drift caused by transistor parameter variations and finite transistor β mismatch. To compensate for this, trimmer resistor R_{13} was introduced to manually null the output offset and restore symmetrical waveform centering around zero volts.

Minor deviations between theoretical and measured responses were caused by:

- transistor gain variations,
- temperature-dependent transistor behavior,
- capacitor leakage currents,
- op-amp offset voltages,
- and parasitic wiring capacitances.

Despite these practical limitations, the circuit produced stable and musically useful decaying drum waveforms with good agreement between theory, simulation, and hardware measurements.

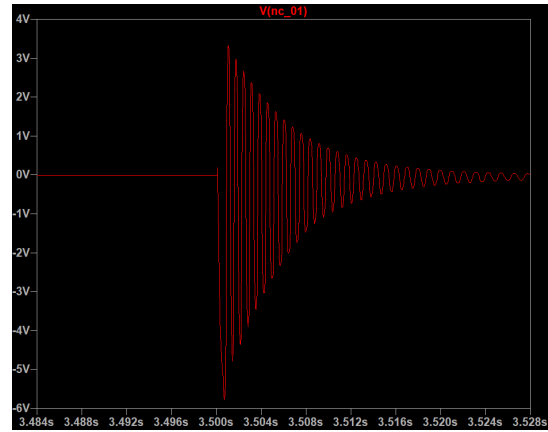


Figure 9: LTspice simulation showing exponentially decaying sinusoidal waveform generated by the VCA and decay envelope circuit.

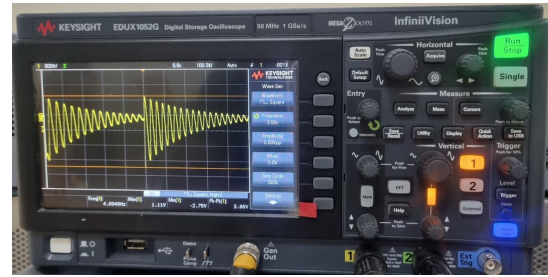


Figure 10: Oscilloscope capture of the practical exponentially decaying drum waveform produced by the implemented VCA circuit.

7. Stage 4: Gate-to-Trigger Converter and Pitch Envelope

The drum synthesizer operates using external clock or gate signals generated from a sequencer or square-wave source. However, directly applying a long-duration gate signal to the envelope circuits creates continuous charging behavior and prevents proper transient drum generation. Therefore, the gate signal must first be converted into a very short trigger pulse.

The purpose of the Gate-to-Trigger converter is to transform an incoming square-wave gate into a narrow transient pulse capable of momentarily activating the envelope circuits. This ensures that every drum hit begins with a clean and well-defined

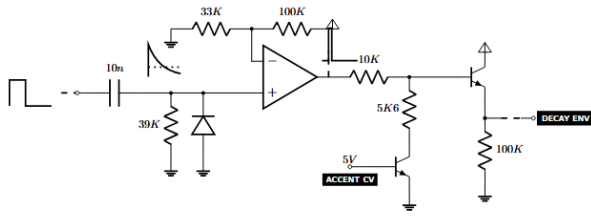


Figure 11: Gate-to-trigger converter circuit.

transient regardless of the duration of the incoming gate signal.

After trigger generation, the signal is used to activate the pitch envelope circuit. The pitch envelope dynamically varies the VCO control voltage over time, producing the characteristic downward frequency sweep commonly found in analog kick drums and tom sounds.

Initially, the oscillator frequency starts at a higher value immediately after triggering. As the envelope capacitor discharges exponentially, the control voltage gradually decreases, causing the oscillator frequency to decay toward its base tuning frequency. This rapid high-to-low pitch sweep is one of the defining acoustic characteristics of analog percussion synthesis.

7.1 Design Rationale

7.2 Mathematical Derivation of the Gate-to-Trigger Converter

The incoming square-wave gate signal is first applied to an RC differentiator network consisting of capacitor C_3 and resistor R_{16} .

The RC network generates an exponentially decaying transient response:

$$V_+(t) = V_0 e^{-t/\tau_{trig}} \quad (38)$$

where:

$$\tau_{trig} = R_{16}C_3 \quad (39)$$

The generated transient is applied to the non-inverting terminal of comparator U4. The inverting terminal is held at a fixed reference voltage:

$$V_- = V_{ref} \quad (40)$$

The comparator output remains HIGH while:

$$V_+(t) > V_{ref} \quad (41)$$

Substituting the transient equation:

$$V_0 e^{-t/\tau_{trig}} > V_{ref} \quad (42)$$

Solving for the trigger duration:

$$e^{-t/\tau_{trig}} > \frac{V_{ref}}{V_0} \quad (43)$$

Taking natural logarithm:

$$-\frac{t}{\tau_{trig}} > \ln \left(\frac{V_{ref}}{V_0} \right) \quad (44)$$

Therefore:

$$T_{trigger} = \tau_{trig} \ln \left(\frac{V_0}{V_{ref}} \right) \quad (45)$$

Using the implemented component values:

$$\tau_{trig} = R_{16}C_3 = 0.39 \text{ ms} \quad (46)$$

The resulting trigger pulse width was experimentally observed to be approximately:

$$T_{trigger} \approx 0.47 \text{ ms} \quad (47)$$

A source follower transistor stage was added at the output to buffer and stabilize the trigger voltage at approximately 5 V. This ensured reliable triggering of the subsequent envelope circuits without loading the comparator output.

Physical Meaning:

The RC network converts a long-duration square-wave gate into a short exponentially decaying pulse. The comparator continuously compares this transient against a fixed threshold volt-

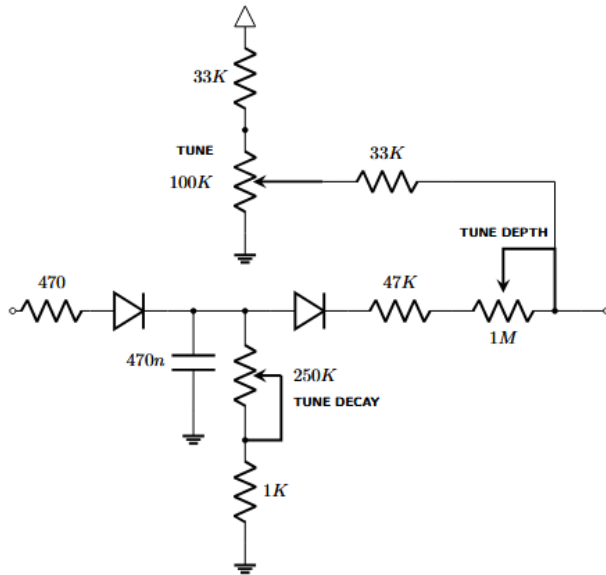


Figure 12: Pitch Envelope Implementation Circuit

age. Once the transient falls below the threshold, the comparator output immediately switches LOW, thereby generating a narrow trigger pulse.

This mechanism ensures that every incoming gate produces only a single short activation pulse independent of the gate duration. Such short transient triggers are essential for percussion synthesis because they allow the envelope generators to restart cleanly for each drum hit.

7.3 Pitch Envelope Derivation

The trigger pulse generated by the Gate-to-Trigger converter is next applied to the pitch envelope circuit.

The VCO already operates at a fixed base tuning voltage:

$$V_{base} \quad (48)$$

When a trigger arrives, capacitor C_4 rapidly charges and then discharges through the RC network and potentiometer control circuitry. This produces an exponentially decaying control voltage:

$$V_{env}(t) = V_0 e^{-t/\tau_{pitch}} \quad (49)$$

The pitch envelope is summed with the base tuning voltage:

$$V_{CV}(t) = V_{base} + V_{env}(t) \quad (50)$$

Including the potentiometer scaling network:

$$V_{CV}(t) = V_{base} + V_0 \cdot \frac{R_{26}}{R_{25} + R_{26}} \cdot e^{-t/\tau_{pitch}} \quad (51)$$

Since the VCO frequency is directly proportional to the control voltage:

$$f(t) \propto V_{CV}(t) \quad (52)$$

the resulting oscillator frequency becomes:

$$f(t) = f_{base} + f_{sweep} e^{-t/\tau_{pitch}} \quad (53)$$

Physical Meaning:

Immediately after triggering ($t = 0$), the pitch envelope voltage is maximum. This causes the oscillator frequency to jump to a higher initial value.

As the envelope capacitor discharges exponentially, the control voltage decreases gradually and the oscillator frequency falls back toward the base tuning frequency.

This rapid downward pitch sweep closely resembles the physical behavior of real percussion instruments. In acoustic drums, the drum membrane initially vibrates with higher tension immediately after impact, producing a higher initial frequency that gradually settles down as the vibration decays.

By varying the RC time constant and potentiometer values, different percussion characteristics such as kicks, toms, and electronic percussion sweeps can be generated.

7.4 Results and Interpretation

The Gate-to-Trigger converter successfully transformed incoming square-wave gate signals into short-duration trigger pulses suitable for activating the envelope generators.

Both LTspice simulations and practical oscilloscope measurements verified that the RC differentiator generated fast transient spikes with rapid exponential decay. The comparator stage accurately detected the transient crossing threshold and produced clean digital trigger pulses.

The generated trigger pulses were sufficiently short to ensure proper re-triggering behavior without causing overlapping envelope activations. The source follower output stage provided stable 5 V trigger levels while preventing loading effects on the comparator.

The pitch envelope circuit successfully produced the intended downward frequency sweep characteristic of analog drum synthesizers. Immediately after triggering, the oscillator frequency increased sharply and then gradually decayed toward the base frequency.

The potentiometer-based control network allowed adjustment of both the pitch sweep depth and decay speed. Faster envelope decay settings produced short punchy kick-like transients, while slower settings generated longer resonant pitch sweeps suitable for tom-style sounds.

The experimentally observed frequency sweep closely matched the theoretical exponential decay model. Minor deviations were caused by:

- capacitor tolerances,
- leakage currents,
- comparator switching delays,
- transistor non-idealities,
- and parasitic wiring capacitances.

Despite these practical limitations, the implemented system produced stable and repeatable trigger generation and pitch modulation behavior with good agreement between theory, simulation, and hardware measurements.

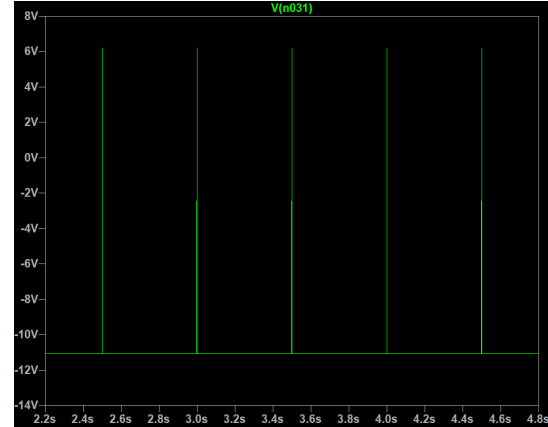


Figure 13: LTspice simulation showing gate-to-trigger conversion and exponentially decaying pitch envelope waveform.

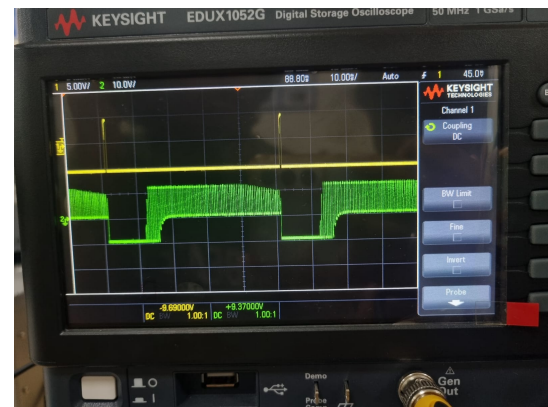


Figure 14: Oscilloscope capture of the practical trigger pulse and pitch envelope response.

8. Complete System Integration and Hardware Implementation

8.1 Complete System Overview

After individually designing and validating each functional stage, the complete analog drum synthesizer system was assembled and tested as an

integrated unit. The final system combines the Voltage Controlled Oscillator (VCO), diode wave shaper, Voltage Controlled Attenuator (VCA), decay envelope generator, gate-to-trigger converter, and pitch envelope circuitry into a single analog percussion synthesis chain.

The complete signal flow of the synthesizer is summarized as follows:

1. An external square-wave or clock signal acts as the trigger source.
2. The Gate-to-Trigger converter transforms the incoming gate into a short transient pulse.
3. The trigger simultaneously activates:
 - the decay envelope controlling amplitude,
 - and the pitch envelope controlling oscillator frequency sweep.
4. The VCO generates a controllable triangular waveform whose frequency varies according to the pitch envelope.
5. The diode wave shaper converts the triangular waveform into a smoother pseudo-sinusoidal waveform.
6. The VCA applies exponential amplitude decay to the waveform using transistor-based analog gain control.
7. The final output produces drum-like analog percussion signals with controllable pitch, decay, attack, and tonal characteristics.

The synthesizer was inspired by the architecture and sound synthesis philosophy of the *Roland TR-909* analog drum machine. While significantly simplified in scale, the implemented system successfully reproduces several important analog percussion synthesis characteristics including:

- transient attack generation,

- exponential amplitude decay,
- pitch sweep modulation,
- analog waveform shaping,
- and resonant drum-like tonal behavior.

The entire project was implemented using discrete analog components including operational amplifiers, BJTs, diodes, passive RC networks, and potentiometer-controlled parameter tuning.

8.2 LTspice Circuit Simulation

Prior to hardware implementation, the complete synthesizer circuit was simulated in LTspice to verify the interaction between all functional stages.

The simulation confirmed:

- stable VCO oscillation,
- proper gate-to-trigger conversion,
- exponential decay envelope generation,
- frequency sweep behavior of the pitch envelope,
- and successful amplitude shaping by the VCA.

The simulated output waveforms closely matched the theoretically expected exponentially decaying drum signals.

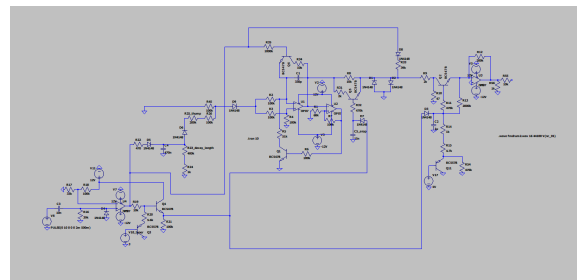


Figure 15: Complete LTspice schematic of the analog drum synthesizer system.

8.3 Hardware Implementation

The complete synthesizer was constructed using discrete analog components on hardware prototyping boards. OP07 operational amplifiers were used for integrator, comparator, and feedback amplification stages, while BC547B transistors were used for the VCA and output buffering sections.

Careful attention was required during hardware implementation to minimize:

- parasitic capacitances,
- grounding noise,
- DC offset drift,
- and trigger instability.

Separate tuning potentiometers were included for:

- oscillator frequency,
- decay time,
- pitch sweep depth,
- and transient response.

This allowed real-time modification of the generated drum sounds and enabled the synthesizer to produce a wide variety of percussion-like tones.

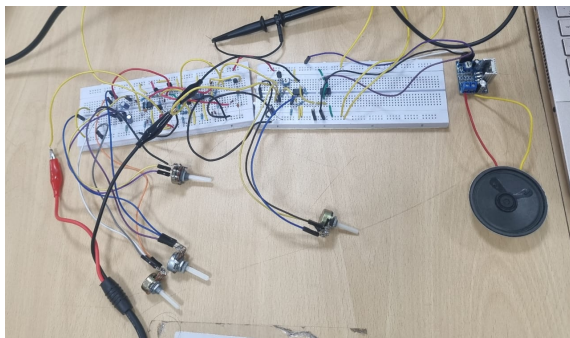


Figure 16: Complete hardware implementation of the analog drum synthesizer.

8.4 Waveform Analysis and Experimental Results

The output waveforms were analyzed using a Keysight EDUX1052G digital storage oscilloscope. The synthesizer was excited using a periodic square-wave trigger signal, and the resulting analog drum waveforms were recorded under different potentiometer settings.

The generated waveforms demonstrated strong agreement with theoretical predictions and LTspice simulations.

Different combinations of frequency tuning, decay control, and pitch envelope depth produced a wide range of percussion characteristics including:

- kick drum tones,
- tom-like resonant sounds,
- transient click sounds,
- short percussive impulses,
- and sustained metallic resonances.

The most important observed feature was the characteristic exponentially decaying sinusoidal waveform accompanied by a transient pitch sweep immediately after triggering. This behavior closely resembles the acoustic response of real percussion instruments.

The transient attack generated by the Gate-to-Trigger converter produced sharp initial excitation, while the VCA ensured natural amplitude decay. Simultaneously, the pitch envelope created the characteristic descending frequency sweep responsible for the “thump” associated with analog kick drums.

Minor deviations between simulation and hardware measurements were caused by:

- component tolerances,

- transistor parameter variations,
- op-amp non-idealities,
- capacitor leakage,
- parasitic effects,
- and breadboard wiring capacitances.

Despite these practical limitations, the complete synthesizer operated reliably and produced stable repeatable analog percussion waveforms.

8.5 LTspice Waveform Simulations

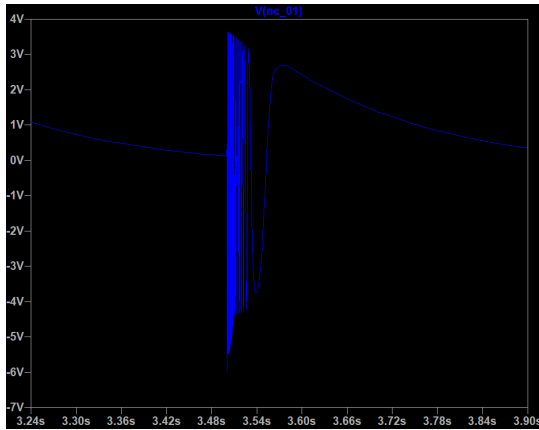


Figure 17: LTspice simulation showing exponentially decaying kick drum waveform.

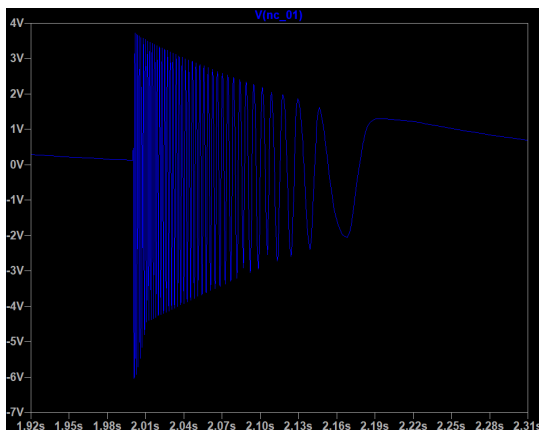


Figure 18: LTspice simulation showing pitch sweep behavior immediately after triggering.

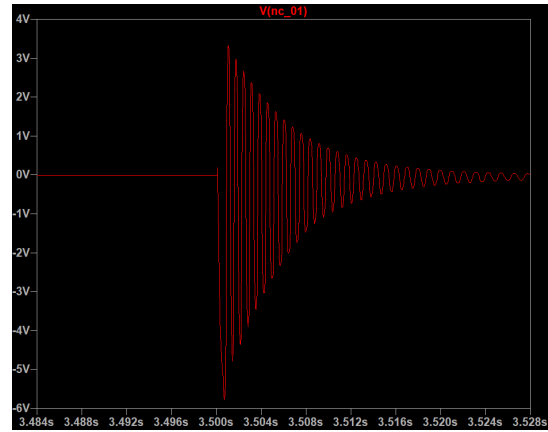


Figure 19: LTspice simulation showing variation in pitch for different potentiometer settings.

8.6 Oscilloscope Measurements

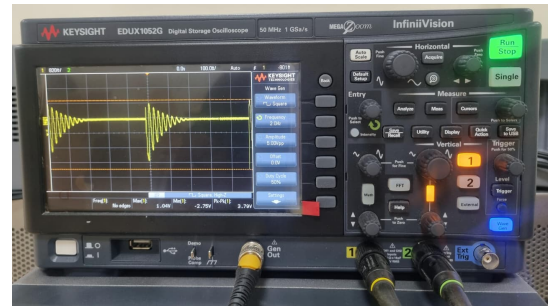


Figure 20: Oscilloscope capture of exponentially decaying kick drum waveform.



Figure 21: Oscilloscope capture showing pitch sweep behavior immediately after triggering.

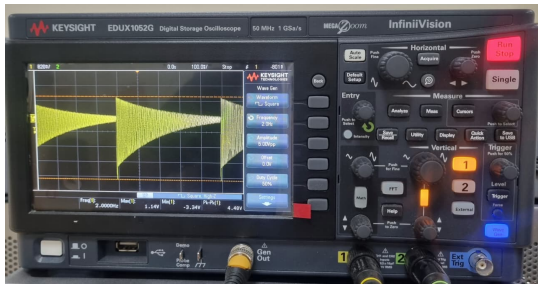


Figure 22: Oscilloscope capture showing variation in pitch for different potentiometer settings.

8.7 Conclusion

The project successfully demonstrated the design and implementation of a fully analog drum synthesizer using discrete electronic components. By combining waveform generation, analog wave shaping, envelope modulation, and transistor-based amplitude control, the system was capable of producing a wide range of percussion-like sounds.

The implemented architecture successfully reproduced several important characteristics of classic analog drum machines, particularly the transient-rich behavior and exponential decay response associated with the Roland TR-909 style synthesis approach.

The VCO provided stable controllable waveform generation, the diode wave shaper reduced harmonic sharpness to produce smoother tonal content, the VCA generated natural exponential decay envelopes, and the pitch envelope introduced realistic transient frequency sweeps.

The experimental results showed strong agreement with theoretical derivations and LTspice simulations. Although practical limitations such as component tolerances and parasitic effects introduced minor deviations, the system remained stable and functionally reliable throughout testing.

Overall, the project demonstrated how complex analog percussion synthesis can be achieved using fundamental electronic building blocks without relying on digital processing or dedicated syn-

thesis ICs. The synthesizer provides an effective educational platform for studying analog sound synthesis, transient response behavior, envelope generation, and voltage-controlled audio processing.

9. System Analysis, Limitations, and Sources of Error

The complete analog drum synthesizer demonstrated stable operation and successfully generated a wide range of percussion-like sounds. However, practical analog hardware systems inherently exhibit several non-ideal effects and limitations that influence overall performance.

9.1 Component Tolerances

One of the primary sources of deviation between theoretical predictions and measured results was component tolerance variation. Since the synthesizer relies heavily on RC timing networks, small variations in resistor and capacitor values directly affected:

- oscillation frequency,
- envelope decay time,
- pitch sweep duration,
- and trigger pulse width.

Electrolytic capacitors particularly exhibited larger tolerance ranges and leakage currents, resulting in slight inconsistencies in envelope timing behavior. Similarly, resistor tolerances caused minor shifts in comparator thresholds and oscillator tuning frequencies.

In multi-channel synthesizer implementations, unmatched components may produce noticeable

differences between identical voice circuits. Therefore, precision resistors and tightly matched capacitors are preferred for improved repeatability and tuning consistency.

9.2 Thermal Effects and Transistor Variations

The transistor-based VCA relied on small-signal BJT transconductance behavior, which is temperature dependent. The transistor transconductance is given by:

$$g_m = \frac{I_C}{V_T} \quad (54)$$

where the thermal voltage V_T varies with temperature.

As ambient temperature changed, transistor gain and base-emitter characteristics shifted slightly, producing small variations in output amplitude and decay response. Additionally, practical BC547B transistors exhibited variations in current gain, causing slight DC offset drift and mismatch between different circuits.

Temperature effects also influenced diode forward voltage thresholds in the wave shaper stage, slightly altering the symmetry and harmonic shaping characteristics of the waveform.

9.3 Parasitic Effects and Breadboard Limitations

Since the hardware implementation was constructed using prototyping boards and discrete wiring, parasitic capacitances and stray inductances were unavoidable.

These parasitic effects introduced:

- slight waveform distortion,
- oscillator frequency drift,
- noise pickup,

- and trigger instability at higher frequencies.

Long breadboard traces increased susceptibility to electromagnetic interference and grounding noise. At higher oscillator frequencies, stray capacitances altered the effective RC time constants and slightly modified the waveform slopes.

Additionally, parasitic capacitances around the comparator stage occasionally introduced trigger timing variations.

A dedicated PCB implementation with proper grounding planes and shorter signal routing would significantly improve overall stability and reduce unwanted parasitic behavior.

9.4 Op-Amp Non-Idealities

Although the OP07 operational amplifier provided stable analog operation, practical non-idealities still affected circuit behavior.

These included:

- finite slew rate,
- limited output swing,
- input offset voltages,
- finite bandwidth,
- and bias currents.

At higher oscillation frequencies, the finite slew rate slightly rounded waveform transitions and introduced deviations from the ideal triangular waveform. Similarly, non-zero op-amp offset voltages contributed to small DC bias errors in the VCA output stage.

The practical output saturation voltage also differed slightly from the ideal supply rails, affecting Schmitt trigger threshold calculations.

9.5 Design Limitations

Although the synthesizer successfully generated analog percussion sounds, the implemented design still possesses several limitations:

- The system generates monophonic drum sounds and does not support simultaneous polyphonic synthesis.
- The wave shaping stage provides only approximate sinusoidal shaping and does not generate a mathematically pure sine wave.
- The pitch envelope and decay envelope are independently adjustable but lack voltage-controlled modulation depth.
- The synthesizer does not include noise generation stages required for realistic snare or hi-hat synthesis.
- Manual tuning of potentiometers is required for sound shaping since no digital control interface is implemented.
- The transistor-based VCA exhibits parameter sensitivity due to transistor mismatch and temperature variation.
- Using precision capacitors and low-tolerance resistors to improve tuning accuracy and envelope consistency.
- Replacing the discrete transistor VCA with OTA-based or IC-based VCAs for improved linearity and thermal stability.
- Adding voltage-controlled modulation inputs for external automation of pitch and decay parameters.
- Incorporating analog noise generators and filter stages to synthesize snare drums, hi-hats, and cymbal sounds.
- Implementing FM or XOR-based metallic synthesis stages similar to advanced analog drum machines.
- Adding MIDI or microcontroller-based triggering for programmable sequencing and synchronization.
- Including active filtering stages for additional tonal shaping and harmonic control.

Despite these limitations, the circuit effectively demonstrates the core principles of analog drum synthesis using simple discrete electronic building blocks.

9.6 Suggestions for Future Improvements

Several improvements can further enhance the performance and flexibility of the synthesizer:

- Designing a dedicated PCB with proper grounding and decoupling to minimize parasitic effects and noise.

These improvements would allow the synthesizer to evolve from a proof-of-concept educational project into a more complete analog percussion synthesis platform.

10. Conclusion

This project successfully demonstrated the design, simulation, and hardware implementation of an analog drum synthesizer using discrete electronic components and analog signal-processing techniques.

The synthesizer was capable of generating a wide variety of percussion-like sounds by controlling waveform frequency, amplitude decay,

transient response, and pitch sweep characteristics. The complete system integrated multiple analog functional blocks including a Voltage Controlled Oscillator (VCO), diode-based wave shaper, transistor-based Voltage Controlled Attenuator (VCA), decay envelope generator, gate-to-trigger converter, and pitch envelope circuitry.

The VCO successfully generated stable triangular waveforms whose frequency varied proportionally with the applied control voltage. The diode wave shaper transformed these waveforms into smoother pseudo-sinusoidal signals suitable for percussion synthesis. The VCA and decay envelope produced exponentially decaying drum waveforms, while the pitch envelope generated characteristic downward frequency sweeps associated with analog kick drums and tom sounds.

Both LTspice simulations and practical hardware oscilloscope measurements showed strong agreement with the theoretical derivations developed throughout the project. Although practical non-idealities such as component tolerances, thermal effects, parasitic capacitances, and transistor parameter variations introduced minor deviations, the overall system remained stable and functionally reliable.

The project also demonstrated how complex analog audio behavior can be achieved using fundamental electronic building blocks without relying on digital synthesis or dedicated audio integrated circuits. In addition to producing musically useful drum sounds, the synthesizer provided valuable insight into analog waveform generation, non-linear signal shaping, RC transient behavior, envelope generation, and voltage-controlled audio processing.

Inspired by the synthesis philosophy of the Roland TR-909 analog drum machine, the implemented system successfully recreated several important characteristics of classic analog percussion

synthesis while remaining compact, modular, and educational in design.

Overall, the project achieved its intended objectives and demonstrated the effectiveness of analog circuit techniques for electronic drum sound synthesis.

Project Demonstration Video:

[https://canva.link/
3jmlirk23hn3ref](https://canva.link/3jmlirk23hn3ref)